Understanding Analysis, 2nd Edition
Chapter 5 - Section 5.2 - Derivatives and the Intermediate Value Property

1. Supply proofs for part (i) and (ii) of Theorem 5.2.4.

Solution: Part (i): $(f+g)'(c) = f'(c) + g'(c)$

Proof:

$$(f+g)'(c) = \lim_{x\to c} \frac{f(x)+g(x)-f(c)-g(c)}{x-c} = \lim_{x\to c}\left(\frac{f(x)-f(c)}{x-c} + \frac{g(x)-g(c)}{x-c}\right)$$
$$= f'(c) + g'(c)$$

where the last step is justified because we know both limits exist, so we can apply the Algebraic Limit Theorem (Theorem 4.2.4).

Part (ii): $(kf)'(c) = kf'(c)$ for all $k \in \mathbf{R}$.

Proof:
$$(kf)'(c) = \lim_{x\to c} k\left(\frac{f(x)-f(c)}{x-c}\right) = k\lim_{x\to c}\frac{f(x)-f(c)}{x-c} = kf'(c)$$

2. Exactly one of the following requests is impossible. Decide which it is, and provide examples for the other three. In each case, assume the functions are defined on all of $\mathbf{R}$.

   (a) Functions f and g not differentiable at zero but where fg is differentiable at zero.

   (b) A function f not differentiable at zero and a function g differentiable at zero where fg is differentiable at zero.

   (c) A function f not differentiable at zero and a function g differentiable at zero where $f+g$ is differentiable at zero.

   (d) A function f differentiable at zero but not differentiable at any other point.

Solution:

(a) $f(x) = 0$ for $x \geq 0$ and $f(x) = 1$ otherwise. $g(x) = 1$ for $x \geq 0$ and $g(x) = 0$ otherwise. Both functions are discontinuous at zero hence not differentiable there, but fg is the zero function.

(b) $f(x) = 0$ for $x \geq 0$ and $f(x) = 1$ otherwise. $g(x) = 0$ everywhere.

(c) Impossible. If $f + g$ were differentiable at zero, then $(f + g) - g = f$ must be differentiable at zero as well by Theorem 5.2.4.

(d) $f(x) = x^2$ for $x \in \mathbf{Q}$ and $f(x) = 0$ otherwise.

It's differentiable at zero because for $c = 0$, $\frac{f(x)-f(c)}{x-c} = x$ and $\lim_{x\to 0} x = 0$.

It's not differentiable at $c \neq 0$. Consider a sequence $(x_n) \to c$. We can split it into two subsequences $(a_n) \in \mathbf{Q}$ and $(b_n) \notin \mathbf{Q}$. We have $(a_n) \to c^2$ and $(b_n) \to 0$. Since $c^2 \neq 0$ for $c \neq 0$, the two subsequences converge to different limits, hence (x_n) doesn't converge.

3. (a) Use Definition 5.2.1 to produce the proper formula for the derivation of $f(x) = 1/x$.

(b) Combine the result in part (a) with the Chain Rule to supply a proof for part (iv) of Theorem 5.2.4.

(c) Supply a direct proof of Theorem 5.2.4 (iv) by algebraically manipulating the difference quotient for (f/g) in a style similar to the proof of Theorem 5.2.4 (iii).

Solution:

(a)
$$\frac{f(x) - f(c)}{x - c} = \frac{\frac{1}{x} - \frac{1}{c}}{x - c} = \frac{-(x - c)}{xc}\frac{1}{x - c} = -\frac{1}{xc}$$

So we have $f'(c) = \lim_{x\to c}(-\frac{1}{xc}) = -\frac{1}{c^2}$ for $c \neq 0$.

(b) We first derive a formula for the derivative of $h(x) = 1/g(x)$.

$$\frac{h(x) - h(c)}{x - c} = \frac{-(g(x) - g(c))}{g(x)g(c)}\frac{1}{x - c} = -\frac{1}{g(x)g(c)} \cdot \frac{g(x) - g(c)}{x - c}$$

Taking the limit to c and noting that the second term of the product is the definition of $g'(c)$

$$h'(c) = \lim_{x\to c} -\frac{1}{g(x)g(c)} \cdot \frac{g(x) - g(c)}{x - c} = -\frac{g'(c)}{g(c)^2}$$

Now apply the product rule to $(fh)'(c)$ to get the formula for $(f/g)'(c)$.

—– Simpler solution by chatgpt —–

Derive the formula of $h'(x)$ by defining $h(x) = 1/x$ then apply the chain rule to $h(g(x))$.

(c)

$$
\begin{aligned}
\frac{\frac{f}{g}(x) - \frac{f}{g}(c)}{x-c} &= \frac{\frac{f}{g}(x) - \frac{f(x)}{g(c)} + \frac{f(x)}{g(c)} - \frac{f}{g}(c)}{x-c} \\
&= \frac{f(x)}{x-c}\left(\frac{g(c)-g(x)}{g(x)g(c)}\right) + \frac{1}{g(c)(x-c)}\left(f(x)-f(c)\right) \\
&= -\frac{f(x)}{g(x)g(c)}\left(\frac{g(x)-g(c)}{x-c}\right) + \frac{1}{g(c)}\left(\frac{f(x)-f(c)}{x-c}\right)
\end{aligned}
$$

Take the limit to c to get

$$-\frac{f(c)g'(c)}{g(c)^2} + \frac{f'(c)}{g(c)} = \frac{f'(c)g(c) - f(c)g'(c)}{g(c)^2}$$

4. Follow these steps to provide a slightly modified proof of the Chain Rule.

   (a) Show that a function $h : A \to \mathbf{R}$ is differentiable at $a \in A$ if and only if there exists a function $l : A \to \mathbf{R}$ which is continuous at a and satisfies

   $$h(x) - h(a) = l(x)(x-a) \qquad \text{for all } x \in A.$$

   (b) Use this criterion for differentiability (in both directions) to prove Theorem 5.2.5.

Solution:

(a) First we prove that if such a function l exists then h is differentiable at a. We have

$$h'(a) = \lim_{x\to a} \frac{h(x)-h(a)}{x-a} = \lim_{x \to a} l(x) = l(a)$$

Next, we prove that if h is differentiable at a then such a function l exists. Define

$$l(x) = \begin{cases} \frac{h(x)-h(a)}{x-a} & x \neq a \\ h'(a) & (x = a) \end{cases}$$

l is continuous at a since $\lim_{x\to a} l(x) = \lim_{x\to a} \frac{h(x)-h(a)}{x-a} = h'(a) = l(a)$.

$h(x) - h(a) = l(x)(x-a)$ for all $x \in A$ by definition.

(b) TODO.

5. Let $l(x) = \begin{cases} x^a & x > 0 \\ 0 & x \leq 0 \end{cases}$.

(a) For which values of a is f continuous at zero?

(b) For which values of a is f differentiable at zero? In this case, is the derivative function continuous?

(c) For which values of a is f twice-differentiable?

Solution:

(a) $a > 0$. The limit at 0 from the left is 0. If $a < 0$ then the right limit diverges. If $a = 0$ then the right limit is 1. But if $a > 0$ then the right limit is 0.

(b) From the direction of positive x, we have $\lim_{x \to 0^+} \frac{x^a - 0^a}{x - 0} = \lim_{x \to 0^+} x^{a-1} = 0^{a-1}$. This is only defined for $a > 1$, in which case it's 0 and agrees with the left limit, so the derivative is defined.

The derivative is continuous, but seems to require us to use the fact that $l'(x) = ax^{a-1}$ for *non-integer* a, which we have not proved yet.

(c) The next right-side derivative at zero is $\lim_{x \to 0^+} x^{a-2} = 0^{a-2}$, so we need $a > 2$.

6. Let g be defined on an interval A and let $c \in A$.

(a) Explain why $g'(c)$ in Definition 5.2.1 could have been given by

$$g'(c) = \lim_{h \to 0} \frac{g(c+h) - g(c)}{h}$$

(b) Assume A is open. If g is differentiable at $c \in A$, show

$$g'(c) = \lim_{h \to 0} \frac{g(c+h) - g(c-h)}{2h}$$

Solution:

(a) Let $h = x - c$. Then we have

$$g'(c) = \lim_{(x-c) \to 0} \frac{g(x) - g(c)}{x - c}$$

And use the fact that $\lim_{(x-c) \to 0} = \lim_{x \to c}$.

(b)

$$\begin{aligned}\lim_{h\to 0}\frac{g(c+h)-g(c-h)}{2h} &= \lim_{h\to 0}\frac{g(c+h)-g(c)+g(c)-g(c-h)}{2h}\\ &= \frac{1}{2}\lim_{h\to 0}\left(\frac{g(c+h)-g(c)}{h}+\frac{g(c)-g(c-h)}{h}\right)\\ &= \frac{1}{2}(g'(c)+g'(c)) = g'(c)\end{aligned}$$

We used the fact that $g'(c) = \lim_{h\to 0}\frac{g(c)-g(c-h)}{h}$, which we'll now prove. Let $h = c - x$. Then we have

$$\lim_{(c-x)\to 0}\frac{g(c)-g(x)}{c-x} = \lim_{(c-x)\to 0}\frac{g(x)-g(c)}{x-c}$$

And use the fact that $\lim_{(c-x)\to 0} = \lim_{(x-c)\to 0} = \lim_{x\to c}$ (this is easy to prove with the usual $\epsilon - \delta$).

One more thing to note is why the problem assumes A is open. This is so that both $g(c-h)$ and $g(c+h)$ are defined.

— **Slight improvement by chatgpt** —
To prove $g'(c) = \lim_{h\to 0}\frac{g(c)-g(c-h)}{h}$, set $k = -h$. Then as $h \to 0$, also $k \to 0$, and

$$\frac{g(c)-g(c-h)}{h} = \frac{g(c+k)-g(c)}{k}$$

Then reuse part (a).

7. TODO

8. TODO

9. True / false with proof / counterexample.

   (a) If f' exists on an interval and is not constant, then f' must take on some irrational values.

   (b) If f' exists on an open interval and there is some point c where $f'(c) > 0$, then there exists a $\delta-$neighborhood $V_\delta(c)$ around c in which $f'(x) > 0$ for all $x \in V_\delta(c)$.

   (c) If f is differentiable on an interval containing zero and if $\lim_{x\to 0} f'(x) = L$, then it must be that $L = f'(0)$.

Solution:

(a) True, by Darboux Theorem and the fact that the irrationals are dense in $\mathbf{R}$.

(b) — **Incorrect** —

True. Otherwise $f'(c)$ is an isolated point and that violates Darboux Theorem.

— **Corrected** —

The above is wrong because Darboux Theorem only guarantees Intermediate Value Property and not continuity (see Section 4.5).

I would not have been able to come up with the counterexample myself. It's

$$f(x) = \begin{cases} x/2 + x^2 \sin(1/x) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

The $x/2$ term makes $f'(0) = \frac{1}{2}$ (compare this with $f(x) = x^2 + \sin(1/x)$ where $f'(0)$ would be 0).

For $x \neq 0$, $f'(x) = \frac{1}{2} - \cos(1/x) + 2x\sin(1/x)$. For any $V_\delta(0)$ there's always an x such that $\cos(1/x) = 1$, which would make $f'(x)$ negative.